K-Means

In [8]:
```
import pandas as pd
import matplotlib.pyplot as plt
dataset=pd.read_csv("Mall_customers.csv")
dataset
```

Out[8]:

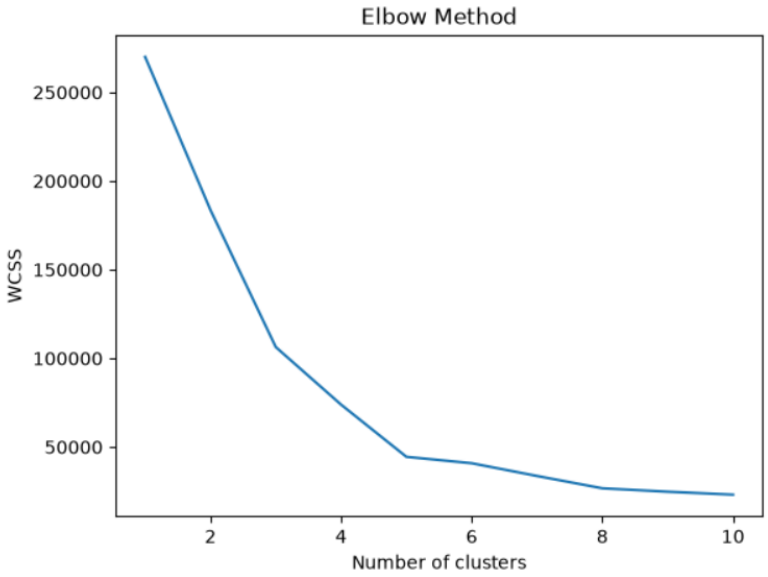
	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [9]:
```
x = dataset.iloc[:,3:5].values
print(x)
```

```
[[ 15  39]
 [ 15  81]
 [ 16   6]
 [ 16  77]
 [ 17  40]
 [ 17  76]
 [ 18   6]
 [ 18  94]
```

In [10]:
```
from sklearn.cluster import KMeans
list = []
for i in range(1,11):
    k = KMeans(n_clusters = i, init = 'k-means++', random_state = 42)
    k.fit(x)
    list.append(k.inertia_)
plt.plot(range(1,11),list)
plt.title('Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()
```



n [11]:
```
print(list)
```

```
[269981.28, 183653.3289473684, 106348.37306211119, 73880.64496247195, 44448.45544793371, 40825.16946386946, 33642.579220779226,
```

n [12]:
```
kmeans=KMeans(n_clusters=5,init='k-means++',random_state=42)
y_means=kmeans.fit_predict(x)
print(y_means)
```

```
[4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4
 2 4 2 4 2 4 0 4 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 1 3 1 0 1 3 1 3 1 0 1 3 1 3 1 3 1 3 1 0 1 3 1 3 1
 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3
 1 3 1 3 1 3 1 3 1 3 1 3 1]
```

In [13]:
```
supervised = dataset
supervised
```

Out[13]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [14]:
```
supervised['cluster_group'] = y_means
supervised
```

Out[14]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	cluster_group
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	196	Female	35	120	79	1
196	197	Female	45	126	28	3
197	198	Male	32	126	74	1
198	199	Male	32	137	18	3
199	200	Male	30	137	83	1

200 rows × 6 columns

In [15]:
```
supervised.to_csv('cluster.csv',index=True)
dir(y_means)
```

Out[15]:
```
['T',
 '__abs__',
 '__add__',
 '__and__',
 '__array__',
 '__array_finalize__',
 '__array_function__',
 '__array_interface__',
 '__array_namespace__',
 '__array_priority__',
 '__array_struct__',
 '__array_ufunc__',
 '__array_wrap__',
 '__bool__',
 '__buffer__',
 '__class__',
 '__class_getitem__',
 '__complex__',
 '__contains__',
 '__copy__',
 '__deepcopy__',
 '__delattr__',
 '__delitem__',
 '__dir__',
 '__divmod__',
 '__dlpack__',
 '__dlpack_device__',
 '__doc__',
 '__eq__',
 '__float__',
 '__floordiv__',
 '__format__',
 '__ge__',
 '__getattribute__',
 '__getitem__',
 '__getstate__',
 '__gt__',
 '__hash__',
 '__iadd__',
 '__iand__',
```

In [16]:
```
centroids=kmeans.cluster_centers_
centroids
```

Out[16]:
```
array([[55.2962963 , 49.51851852],
       [86.53846154, 82.12820513],
       [25.72727273, 79.36363636],
       [88.2       , 17.11428571],
       [26.30434783, 20.91304348]])
```

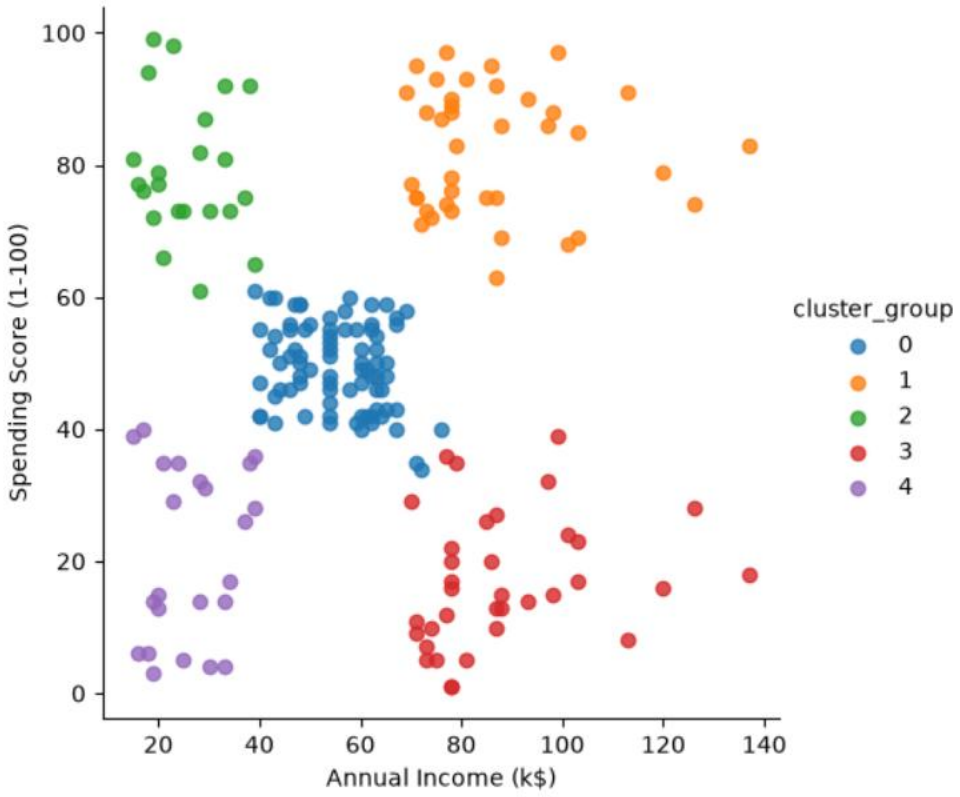
In [17]:
```
supervised.columns[3]
```

Out[17]:
```
'Annual Income (k$)'
```

In [19]:
```
supervised.columns
```

Out[19]:
```
Index(['CustomerID', 'Genre', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)', 'cluster_group'],
      dtype='str')
```

In [18]:
```
import seaborn as sns
facet = sns.lmplot(data=supervised, x=supervised.columns[3],
                   y=supervised.columns[4], hue=supervised.columns[5],
                   fit_reg=False,legend_out=True)
```



Algomerative

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

In [2]:
```
dataset = pd.read_csv('Mall_Customers.csv')
dataset
```

Out[2]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

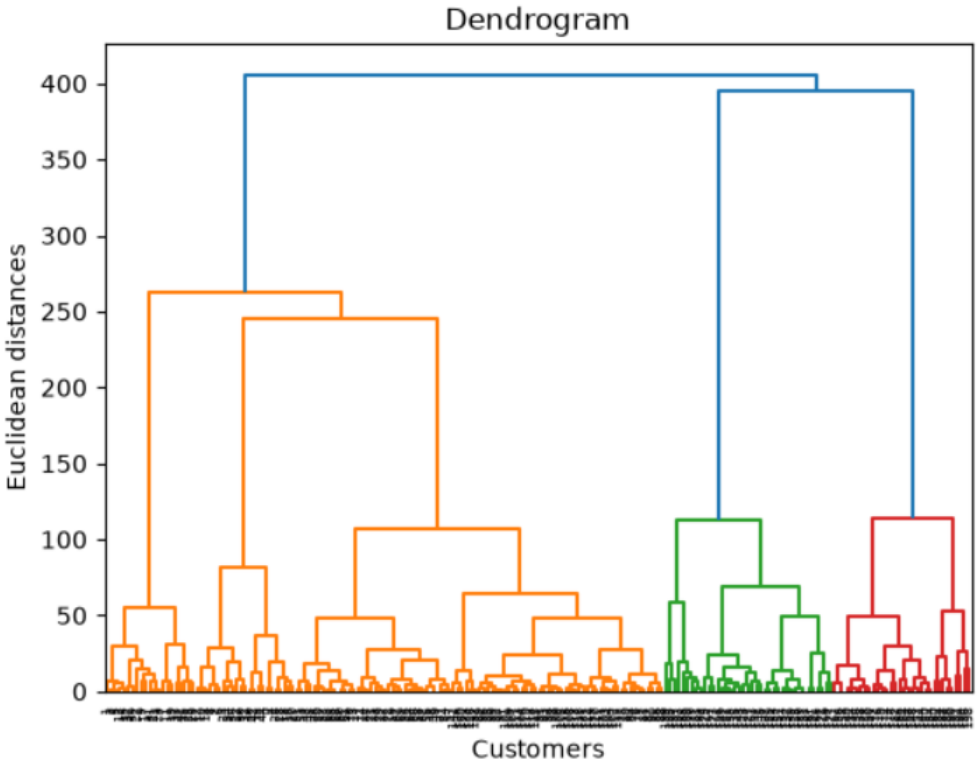
In [3]:
```
X = dataset.iloc[:, 3:5].values
X
```

In [4]:

```
import scipy.cluster.hierarchy as sch
import matplotlib.pyplot as plt

dendrogram = sch.dendrogram(
    sch.linkage(X, method='ward')
)

plt.title('Dendrogram')
plt.xlabel('Customers')
plt.ylabel('Euclidean distances')
plt.show()
```



In [5]:
```
from sklearn.cluster import AgglomerativeClustering

# Create the Agglomerative Clustering model
clusmodel = AgglomerativeClustering(n_clusters=5)

# Fit the model and predict the cluster for each customer
label = clusmodel.fit_predict(X)
label
```

Out[5]:
```
array([4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3,
       4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 1,
       4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 2, 1, 2, 0, 2, 0, 2,
       1, 2, 0, 2, 0, 2, 0, 2, 0, 2, 1, 2, 0, 2, 1, 2, 0, 2, 0, 2, 0, 2,
       0, 2, 0, 2, 0, 2, 1, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2,
       0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2,
       0, 2])
```

In [6]:
```
# Create a new DataFrame from the original dataset
supervised = pd.DataFrame(dataset)
# Add the cluster labels as a new column
supervised['Cluster_group'] = label
# Display the final dataset
supervised
```

Out[6]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	Cluster_group
0	1	Male	19	15	39	4
1	2	Male	21	15	81	3
2	3	Female	20	16	6	4
3	4	Female	23	16	77	3
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	196	Female	35	120	79	2

In [7]:
```
import seaborn as sns
facet = sns.lmplot(data=supervised, x=supervised.columns[3], y=supervised.columns[4], hue=supervised.columns[5],
                   fit_reg=False,legend_out=True)
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\seaborn\regression.py:598: UserWarning: legend_out is deprecated from the
ws`.
  warnings.warn(msg, UserWarning)
```

